MILITARY SPECIFICATION

LUBRICANT, COLLOIDAL GRAPHITE IN OIL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers colloidal graphite in oil lubricant primarily for ordnance but may be used for other purposes (see 6.1).

1.2 Classification. The lubricant covered by this specification shall be of the following grades, as specified (see 6.2):

Grade A - Light

Grade B - Medium

Grade C - Heavy

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-B-585	Box, Wood, Wirebound
PPP-B-621	Box, Wood, Nailed and Lock-Corner

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640, by using The self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

PPP-B-640	Box, Fiberboard, Corrugated, Triple Wall
PPP-P-704	Pails, Metal: (Shipping, Steel, 1 through 12 Gallons)

STANDARDS

FEDERAL

FED-STD-791	Lubricant, Liquid Fuels, and Related Products; Methods of Testing

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290	Packaging, Packing and Marking of Petroleum and Related Products
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D96	Water and Sediment in Crude Oils
D97	Pour Point of Petroleum Oils

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D270	Sampling Petroleum and Petroleum Products
D445	Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)
D3278-78	Standard Method of Test for Flash Points of Liquid by Set and Flash Closed Tester
D2422	Viscosity System for Industrial Fluid Lubricants

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The lubricants shall consist of stabilized colloidal electric furnace graphite dispersed in refined mineral lubricating oils. Solids other than stabilized colloidal graphite shall not be used.

3.2 Physical and chemical requirements. The colloidal graphite in oil lubricants shall conform to the physical and chemical requirements shown in TABLE I.

TABLE I. Physical and chemical requirements.

	Grade A	Grade B	Grade C	Test Paragraph
Viscosity $\left\{cSt\left[\frac{mm^2}{s}\right]\right\}$ at 4 °C	90-110	---	---	4.5.1
Viscosity $\left\{cSt\left[\frac{mm^2}{s}\right]\right\}$ at 99°C	--	22-27	57 (min)	4.5.1
Flash point °C (min.)	140	160	160	4.5.2
Pour point °C (max.)	-46	-9	-9	4.5.3
Solids content, weight percent	2.00± 0.05	10.00± 0.25	10.00± 0.25	4.5.4
Ash, weight percent, (max.)	.75	.75	.75	4.5.5
Hiding power of lubricant containing 2 weight percent solids sq. ft. per gallon (min.)	1,200	1,200	900	4.5.7

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3.3 Stability.

3.3.1 Grades A and B. When Grades A and B lubricants are tested as specified in 4.5.6, not less than 95 weight percent of the solids shall be in suspension after centrifuging.

3.3.2 Grade C. When Grade C lubricant is tested as specified in 4.5.6, not less than 85 weight percent of the solids shall be in suspension after centrifuging.

3.4 Particle size, Grades A and B. When subjected to microscopic examination as specified in 4.5.8, over 50% of the graphite particles shall be smaller than 1 micron (in largest dimension). The remainder shall be smaller than 4 microns.

3.5 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering this class of material and the final product shall be homogeneous and free from lumps and foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot formation. A lot shall consist of an indefinite number of unit containers of identical size and type, offered for acceptance, and filled with a homogeneous mixture of material from one isolated container; or filled with a homogeneous mixture of material manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material. In the event the process is a batch operation, each batch constitutes a lot (see 6.3).

4.3 Sampling.

4.3.1 Sampling for examination of packaging. A random sample of packaged containers shall be taken from each lot in accordance with MIL-STD-105 at inspection level II and acceptable quality level of 2.5 percent defective. The sample shall be examined in accordance with 4.4.1

4.3.2 Sampling for tests. The sample for tests shall be taken in accordance with ASTM D 270 from each lot of lubricant offered or processed for delivery under the contract or order.

4.4 Quality conformance inspection. Inspection shall be performed in accordance with method 9601 of FED-STD-791.

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4.4.1 Examination of the packaging. Samples taken in accordance with 4.3.1 shall be examined for compliance with Section 5 requirements with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Any container having one or more defects or having less than the required fill shall be rejected. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.4.2 Tests. Samples of the product offered for acceptance, obtained in accordance with 4.3.2, shall be tested to determine conformance to all the requirements specified in Section 3.

4.4.3 Workmanship. Each container of fluid shall be subject to examination. Failure to comply with 3.5 shall be cause for rejection.

4.5 Methods of inspection.

4.5.1 Viscosity. The viscosity shall be determined in accordance with ASTM D445.

4.5.2 Flash point. The flash point shall be determined in accordance with ASTM D3278-78.

4.5.3 Pour point. The pour point shall be determined in accordance with ASTM D97.

4.5.4 Solids content. The solids content shall be determined by flocculating the solids in a weighed sample with ethyl acetate, filtering through a tared Gooch crucible, washing and weighing.

4.5.4.1 Preparation of Gooch crucible. Acid washed asbestos shall be suspended in water by the usual method of vigorous shaking or mechanical stirring. An asbestos pad shall be built up in the crucible to a thickness of approximately 1/4 to 3/8 inch by pouring two or three portions of the asbestos water suspension into the crucible, without tamping or packing the pad. The crucible shall then be dried at 110°C for 3 hours, cooled in a desiccator for 1/2 hour, and weighed.

WARNING

Personnel required to handle dry asbestos shall be adequately protected so as not to exceed asbestos exposure limits established by 29 CFR 1910.100(b)(2).

4.5.4.2 Test procedure. Accurately weigh a sample of approximately 0.5 g of the lubricant into a 50 ml beaker. To flocculate the graphite, dilute the sample with 10 ml of chloroform and stir with a glass rod until uniform, then add approximately 20 ml of ethyl acetate and again stir. Within 15 minutes, pour the sample through a tared Gooch crucible under slight vacuum, and carefully wash the graphite from the beaker into the crucible with ethyl acetate, using a rubber policeman to facilitate the operation. Wash the crucible with an additional 50 ml of ethyl acetate. To guard against peptization, place crucible in drying oven at 110°C for 1/2 hour. After drying cool the crucible for a few minutes and wash with 100 ml of technical grade ethyl ether. Again dry crucible at 110°C for 1/2 hour and cool. Next, pour approximately 10 ml of ethyl acetate into crucible, applying little or no vacuum. As the ethyl acetate begins to

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drop from the bottom of the crucible, gradually add chloroform and continue to wash with this solvent. Care must be taken at this point not to wash graphite through crucible. Continue to wash with chloroform until approximately 50 ml has been used. Then dry the crucible as before. After cooling, repeat the washing with chloroform, following the above precautions. Again dry for 1/2 hour, cool for 1/2 hour in desiccator, and weigh. Repeat the chloroform washing treatment, weighing after each treatment, until the weight of the crucible does not decrease. The weight of solids contained in the original weight of sample is the increase in weight of the crucible. Calculate the precent solids content of the lubricant as follows:

$$\text{Percent solids} = \frac{\text{weight solids}}{\text{weight sample}} \times 100$$

4.5.5 Ash. The ash shall be determined by burning a known amount of the lubricant, igniting the residue and weighing the ash.

4.5.5.1 Test procedure. Weigh 10.00 ± 0.01 g of the lubricant into a tared porcelain crucible (size 1A). Support the crucible in a nichrome triangle and warm with a grid type burner. Heat the sides of the crucible above the level of the liquid to prevent creeping of the oil. Then slowly warm the sides of the crucible until the oil ignites and boils gently. Direct the flame across the top and against the side of the crucible, keeping the oil burning gently until the flame dies away. This operation should take at least 10 minutes, and should be carried out in a fume hood. After burning the oil, ignite the crucible with the full heat of the burner until the carbon deposit and the graphite are completely oxidized. Then cool the crucible in a desiccator and weigh. The increase in weight of the crucible is the ash of the lubricant, and this weight, multiplied by the factor 10, gives the percent of ash.

4.5.6 Stability.

4.5.6.1 Grade A. The stability test of Grade A lubricant shall be determined on a representative sample of the product which has been thoroughly stirred to ensure uniformity. From this sample, remove and analyze two portions in accordance with 4.5.4. From the remainder of the sample, take the portions specified and centrifuge as specified in 4.5.6.3 and 4.5.6.4.

4.5.6.2 Grades B and C. The stability test of Grades B and C lubricants shall be conducted on a blend of one part of the lubricant with 49 parts of the neutral refined mineral oil specified in 4.5.7. Heat a 98-gram portion of the neutral refined mineral oil to 66± 5°C and blend thoroughly with the two grams of the lubricant. Determine the solids content of this blend by the technique specified in 4.5.4. Two samples of the blend weighing approximately 15 g each should be analyzed. The results of this analysis gives the initial solids contents of the material to be centrifuged. The blended samples shall be centrifuged as specified in 4.5.6.3 and 4.5.6.4.

4.5.6.3 Centrifuge. The centrifuging shall be conducted in a machine which allows the tubes to swing in a horizontal position at a speed of 2,000 ± 50 r.p.m. with a diameter of swing (tip to tip of whirling tubes) of 14 to 17 inches. The centrifuge head shall carry centrifuge tubes of 15 ml capacity with tapered bottoms. Centrifuging shall be conducted for a period of 11 minutes. If the avialable centrifuge does not meet the limits specified, the proper speed shall be calculated in accordance with ASTM D96.

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4.5.6.4 Test procedure. Two centrifuge tubes shall be filled with the samples prepared as specified in 4.5.6.1 and 4.5.6.2 to a depth of 4 inches and adjusted to 21°C. The tubes are then centrifuged under the conditions specified in 4.5.6.3. After this treatment, decant the supernatant liquid from the tubes into tared beakers, pouring out only that which flows quickly, the whole operation not to require more than 30 seconds. Weigh the decanted material and analyze in accordance with 4.5.4. Calculate the percent solids in suspension after centrifuging as follows:

Grade A:

Percent solids in suspension after centrifuging =

$$\frac{\text{percent solids after centrifuging}}{\text{percent solids before centrifuging}} \times 100$$

Grades B and C:

Percent solids in suspension after centrifuging =

$$\frac{\text{percent solids after centrifuging}}{\text{percent solids in blend before centrifuging}} \times 100$$

4.5.7 Hiding power. The hiding power shall be determined using the Pfund Cryptometer, following the directions of the manufacturer of the instrument. The plate with a wedge constant of 0.0002 shall be used. Since Grades B and C are too opaque for examination, they are blended with a neutral refined petroleum oil of the following characteristics:

Relative density	.9100-.8984
Flash point	202°C (min.)
ISO Viscosity Grade	68
Viscosity, cSt at 100°C	7-8

4.5.7.1 Test procedure. A 2.00 ± 0.01 gm. portion of the Grade B or C lubricant is diluted to 10.00 ± 0.01 gm. with the oil specified in 4.5.7 and the mixture thoroughly blended at a temperature of approximately 66°C. This blend, containing 2.0 percent solids, shall be examined in the Pfund Cryptometer.

4.5.8 Particle size. The particle size shall be determined on Grade A lubricant as received; and on Grade B, by blending one part of Grade B lubricant with 49 parts of the oil specified in 4.5.7. The samples shall be thoroughly stirred to insure uniformity.

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4.5.8.1 Test procedure. A drop of the sample prepared as specified in 4.5.8, is placed on a glass microscopic slide using the utmost care to avoid contamination of the product, slides, or cover glass, with foreign material picked up in normal surroundings. The particle size is determined by taking a photomicrograph of the particles followed by measurement on the print, or by direct measurement on the ground glass screen of a metallographic microscope. A magnification of at least 600 diameters shall be used.

5. PACKAGING

5.1 Packaging. Unless otherwise specified (see 6.2), the colloidal graphite lubricant shall be packaged in 1-quart, 1-gallon, or 5-gallon cans, or 5-gallon closed-top steel drums, conforming to PPP-C-96 and PPP-P-704.

5.2 Packing.

5.2.1 Level A. Packing shall be in accordance with requirements for exterior packing of PPP-C-96 and PPP-P-704, except that only nailed wood or wirebound boxes, conforming to PPP-B-621 or PPP-B-585, respectively, shall be used.

5.2.1.1 Drums. Five-gallon drums will require no further packing.

5.2.2 Level B. Twenty-four 1-quart, six 1-gallon, or two 5-gallon cans shall be packed in a nailed wood or wirebound box conforming to PPP-B-621 or PPP-B-585, respectively; or twenty-four 1-quart or six 1-gallon cans, but not 5-gallon cans, shall be packed in a corrugated or solid fiberboard box conforming to PPP-B-640. Arrangement within the container shall be in accordance with requirements for exterior packing of PPP-C-96 and PPP-P-704.

5.2.2.1 Drums. Five-gallon drums will require no further packing.

5.2.3 Industrial. Containers shall be packed in accordance with the requirements of MIL-STD-1188.

5.3 Marking. Unless otherwise specified in the contract or order, marking for shipping and storage shall be in accordance with MIL-STD-290.

6. NOTES

6.1 Intended use.

6.1.1 Grade A. Grade A lubricant is suitable for the lubrication of machine gun housing guides, windshield wipers, and other lightly loaded sliding members exposed to the weather for both low and elevated temperatures.

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6.1.2 Grade B. Grade B lubricant is suitable for the lubrication of gear trains of torpedoes.

6.1.3 Grade C. Grade C lubricant is suitable for the lubrication of medium or heavy duty gun slides without causing excessive resistance to counterrecoil at ambient temperatures down to -23°C. It should retain sufficient lubricating properties to permit free recoil and counterrecoil when the gun is heated as a result of sustained fire.

6.2 Ordering data. Requests, requisitions, schedules and contracts or orders should specify the following:

a. Title, number, and date of this specification.

b. Grade of lubricant required (see 1.2).

c. Type and size of container in which the lubricant is to be furnished (see 5.1).

d. Selection of levels of protection (see 5.2).

6.3 Definition.

6.3.1 Batch. A quantity of material which has been manufactured by some chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
ARMY - ME
NAVY - OS
AIR FORCE - 68

Preparing Activity:
NAVY - OS

Project Number:
9150-0589

☆U.S. GOVERNMENT PRINTING OFFICE: 1983 — 705-040/5216

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